→ Bit-Flip Channel

$E_0 = \sqrt{p}\, I \qquad E_1 = \sqrt{1-p}\, X$

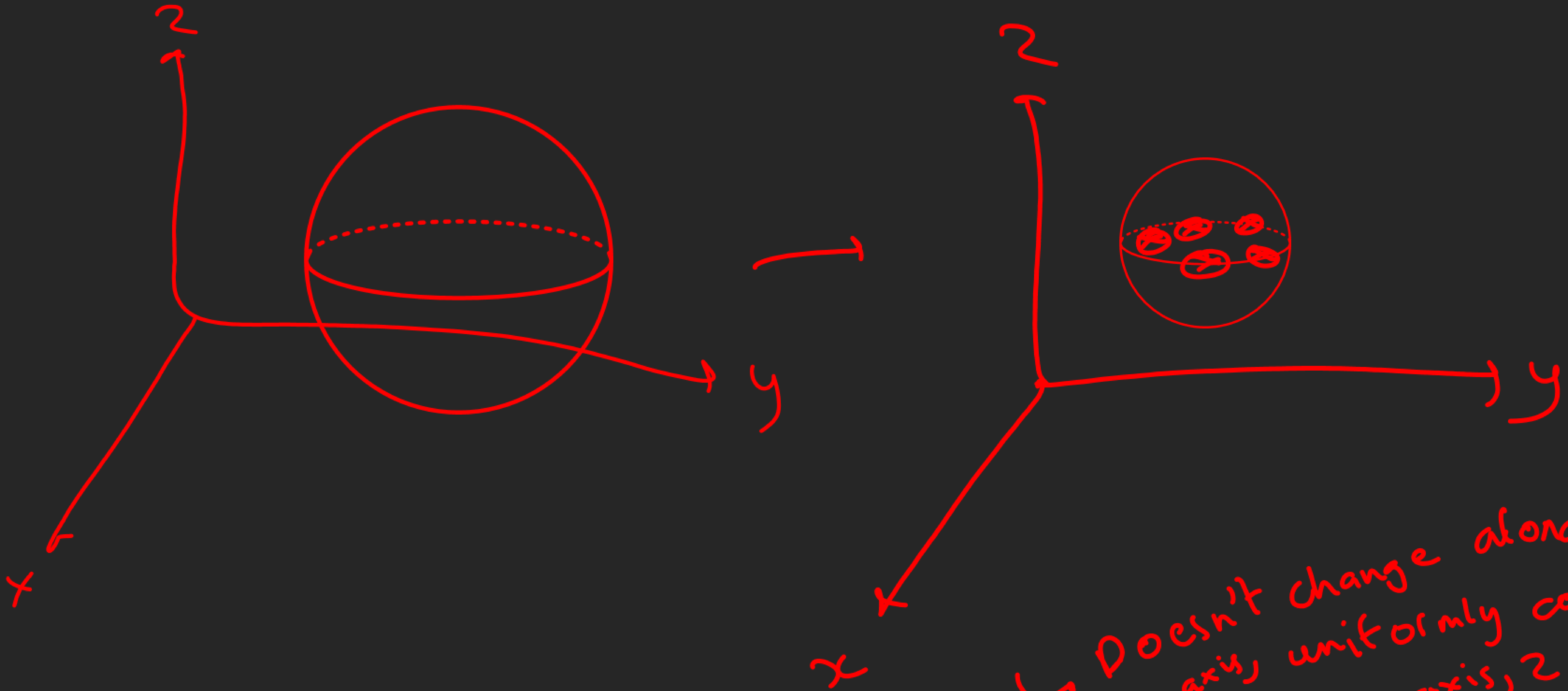
$$\mathcal{E}(\rho) = p\rho + (1-p) X \rho X$$

$$tr(\rho^2) = \frac{1+|r|^2}{2}$$

$|r| \to$ Norm of Bloch vector

↓

Can only decrease from Bit-Flip Channel → Bloch sphere contracts



↳ Doesn't change along x-axis, uniformly contracts along y-axis, z-axis

→ Phase - Flip Channel

$E_0 = \sqrt{p}\ I \qquad E_1 = \sqrt{1-p}\ Z$

$\mathcal{E}(\rho) = p\rho + (1-p) Z \rho Z$

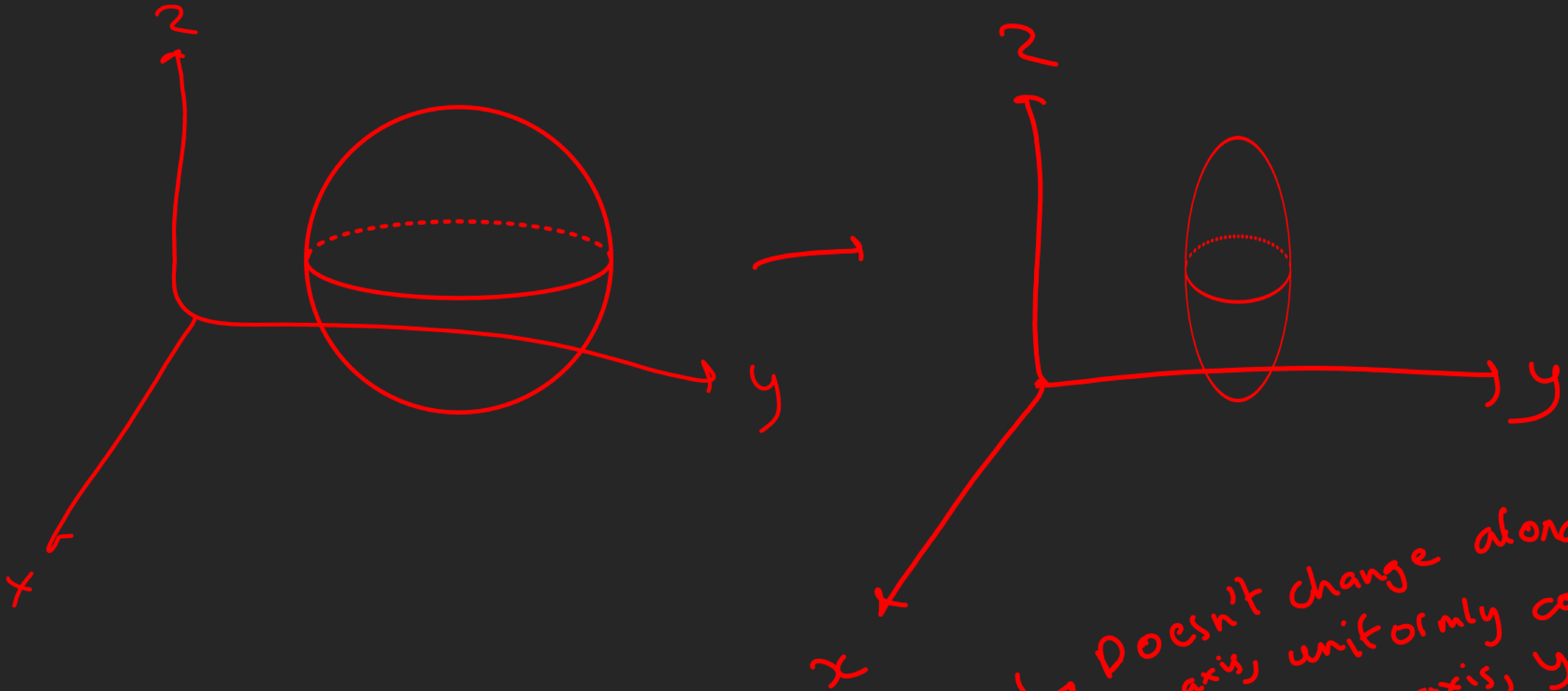
for $p = \frac{1}{2}$ → $\mathcal{E}(\rho) = P_0 \rho P_0 + P_1 \rho P_1$ → Measurement in $(|0\rangle, |1\rangle)$ basis

↓

$(r_x, r_y, r_z) \rightarrow (0, 0, r_z)$

↓

Projected along z - axis



Doesn't change along z-axis, uniformly contracts along x-axis, y-axis

Bit-Phase Flip Channel

$E_0 = \sqrt{p}\, I$
$E_1 = \sqrt{1-p}\, Y$

$$\mathcal{E}(\rho) = p\rho + (1-p) Y \rho Y$$